

LAPPEENRANTA HIEKKAPAKKA, Finland

WMO: 029190

Lat: 61.198N

Lon: 28.473E

Elev: 77

StdP: 100.40

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-23.9	-30.2	0.2	-27.4	-26.6	0.3	-23.9	13.0	-1.2	12.2	-1.9	1.1	280	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	25.8	18.6	24.0	17.8	22.4	16.8	19.7	23.8	18.6	22.8	17.5	21.4	3.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.3	22.2	17.0	12.2	20.7	15.8	11.3	19.6	56.8	23.7	53.1	22.9	49.6	21.4	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.5	9.5	-28.5	27.9	5.8	2.5	-32.6	29.7	-36.0	31.2	-39.3	32.6	-43.5	34.4		
			DB	WB	-28.6	20.6	5.8	1.4	-32.8	21.6	-36.1	22.4	-39.4	23.2	-43.6	24.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-8.1	-8.2	-3.5	1.9	8.4	14.6	16.8	15.7	10.7	4.5	-1.6	-4.7
	DBStd	10.20	6.94	7.73	4.88	4.21	5.04	3.64	3.47	3.47	3.96	3.75	4.92	6.69
	HDD10.0	2852	560	511	420	244	92	4	1	3	39	172	349	457
	HDD18.3	5299	818	744	678	493	307	121	73	94	228	428	599	715
	CDD10.0	643	0	0	0	1	43	143	211	180	62	2	0	0
	CDD18.3	48	0	0	0	0	1	10	25	13	0	0	0	0
	CDH23.3	224	0	0	0	0	5	57	122	41	0	0	0	0
	CDH26.7	26	0	0	0	0	0	11	14	1	0	0	0	0
Wind	WSAvg	4.4	4.1	4.0	3.9	3.6	3.8	4.5	3.9	4.4	4.6	5.3	5.0	5.1
Precipitation	PrecAvg	640	49	38	35	30	42	61	75	74	58	64	60	54
	PrecMax	793	80	91	66	63	141	91	148	135	131	147	127	114
	PrecMin	492	11	4	7	10	7	19	17	16	19	15	9	21
	PrecStd	72	18	20	16	13	29	19	31	35	33	29	25	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	3.2	6.8	17.1	23.3	28.7	28.4	26.4	21.0	13.0	8.7	6.7
		MCWB	1.7	2.0	2.7	10.6	15.7	18.5	19.6	18.9	16.0	10.8	7.6	5.7
	2%	DB	1.8	2.2	4.6	12.6	20.9	24.7	26.3	24.4	18.5	11.6	7.1	4.4
		MCWB	1.3	1.5	1.7	7.2	13.9	17.4	18.9	18.0	14.8	10.2	6.6	3.6
	5%	DB	0.9	1.2	3.1	10.0	18.6	22.7	24.6	22.7	17.3	10.7	5.6	2.9
		MCWB	0.5	0.7	1.0	5.8	12.7	16.2	18.5	17.1	14.3	9.5	5.1	2.3
	10%	DB	0.2	0.6	2.1	7.7	16.3	20.6	22.7	21.1	16.0	9.6	4.4	1.8
		MCWB	-0.1	0.1	0.5	4.4	11.5	15.4	17.6	16.6	13.8	8.5	3.9	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.4	3.5	10.6	16.9	19.6	21.4	20.7	16.9	11.4	7.7	6.0
		MCDB	2.4	2.8	5.5	16.7	22.3	25.3	25.4	24.1	19.2	12.4	8.4	6.7
	2%	WB	1.3	1.7	2.3	7.9	15.0	18.3	20.3	19.2	15.7	10.6	6.6	3.6
		MCDB	1.6	2.1	3.7	11.4	18.8	22.9	24.4	22.8	17.4	11.4	7.1	4.3
	5%	WB	0.6	0.9	1.5	6.2	13.7	17.2	19.2	17.9	14.9	9.8	5.2	2.3
		MCDB	0.9	1.1	2.7	9.4	17.4	21.4	23.4	21.5	16.6	10.6	5.5	2.7
	10%	WB	0.0	0.2	0.8	4.7	11.9	16.0	18.1	16.8	14.0	8.7	3.9	1.4
		MCDB	0.3	0.6	1.8	7.4	16.0	20.0	22.0	20.2	15.8	9.5	4.4	1.8
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.3	7.2	7.6	7.7	7.3	7.0	6.8	5.2	3.6	3.6	4.3
		MCDBR	3.9	4.3	7.6	10.6	11.3	10.7	9.4	9.4	6.4	3.9	3.6	3.1
		MCWBR	3.9	4.0	5.6	6.4	6.0	5.2	4.2	4.3	3.7	3.1	3.5	3.1
	5% WB	MCDBR	3.8	3.4	5.6	9.7	9.3	8.8	7.8	6.5	5.6	3.7	3.7	3.0
		MCWBR	3.8	3.3	4.4	6.2	5.2	4.9	4.0	3.5	3.5	3.2	3.7	3.1
Clear-Sky Solar Irradiance	taub	0.255	0.298	0.324	0.363	0.358	0.355	0.380	0.366	0.342	0.324	0.278	N/A	
	taud	2.066	2.124	2.200	2.336	2.431	2.470	2.418	2.444	2.502	2.401	2.155	N/A	
	Ebn at Noon	452	660	785	829	865	874	838	820	779	649	421	N/A	
	Edh at Noon	40	73	95	102	100	98	102	91	73	57	37	N/A	
All-Sky Solar Radiation	RadAvg	0.24	0.80	2.26	3.68	5.05	5.31	5.19	4.00	2.33	0.94	0.25	0.12	
	RadStd	0.04	0.19	0.31	0.40	0.59	0.35	0.49	0.47	0.26	0.13	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-190	-189	N/A	N/A

Nomenclature: See separate page